

Maths

Assignment (3)

Name - Aayush Agarwal

Class - 2D

Roll No. - (2)

Univ - 12515500008

Age	10-20	20-30	30-40	40-50	50-60	60-70
No of members	3	61	12	15	14	3

Range	x_i	f_i	fix_i	fix_i^2
10-20	15	3	45	675
20-30	25	61	1525	38125
30-40	35	12	420	14700
40-50	45	15	675	30375
50-60	55	14	770	42350
60-70	65	3	195	12675
Σ		108	3630	138900

$$\sigma^2 = \frac{\Sigma fix_i^2}{\Sigma f_i} - \bar{x}^2$$

$$= \frac{138900}{108} - 1129.66 = \frac{138900 - 122004.3}{108}$$

$$= 154.44$$

$$\bar{x} = \frac{\Sigma fix_i}{\Sigma f_i} = \frac{3630}{108} = 33.61$$

$$\sigma^2 = 154.44$$

$$\sigma = 12.50$$

Q2

Interval	f_i	cf	$f_i x_i$
10-20	5	5	75
20-30	f_1	$5+f_1$	$25f_1$
30-40	20	$25+f_1$	700
40-50	f_2	$25+f_1+f_2$	$45f_2$
50-60	5	$30+f_1+f_2$	275

Mean = 35

$$30 + f_1 + f_2 = 60$$

$$f_1 + f_2 = 30$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{5 + f_1 + 20 + f_2 + 5}{60} = 35$$

$$30 + f_1 + f_2 = 2100$$

$$f_1 + f_2 = 2070$$

$$35 = \frac{1050 + 25f_1 + 45f_2}{60}$$

$$2100 - 1050 = 25f_1 + 45f_2$$

$$1050 = 25f_1 + 45f_2$$

$$210 = 5f_1 + 9f_2$$

$$150 = 5f_1 + 5f_2$$

$$\begin{array}{r} 210 \\ - 150 \\ \hline 60 \end{array} = 4f_2$$

$$\frac{60}{4} = f_2$$

$$15 = f_2$$

$$15 = f_1$$

Q3

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

PMF

$X =$ sum of two dice

$$2 = 1/36$$

$$3 = 2/36$$

$$4 = 3/36$$

$$5 = 4/36$$

$$6 = 5/36$$

$$7 = 6/36$$

$$8 = 5/36$$

$$9 = 4/36$$

$$10 = 3/36$$

$$11 = 2/36$$

$$12 = 1/36$$

Q4

0, 0, 2, 2, 2, 3, 4, 5, 6, 7, 8, 9, 15, 18, 20, 25, 26, 27, 27

0, 0, 2, 2, 2, 3, 4, 5, 6, 7, 7, 7, 8, 9, 15, 18, 20, 25, 26, 27

~~10, 19~~

$\therefore Q_2 = \frac{\cancel{19} + \cancel{1}}{2} = \cancel{10}$

a) $Q_2 = 7$

$Q_1 = 2.5$

$Q_3 = 18.5$

$IGR = Q_3 - Q_1$

$= 18.5 - 2.5$

$= 16$

b) $UF = Q_3 + 1.5 IGR$

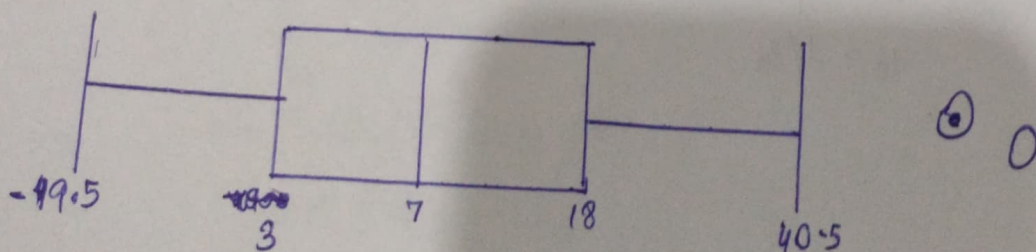
$= 18.5 + 1.5 \times 16$

$= 43.5$

$LF = Q_1 - 1.5 IGR$

$= 2.5 - 1.5 \times 16$

$= -22.5$



No outliers

- Q5
- Ratio
 - Ratio
 - Interval
 - Ordinal

Q6

A die is rolled :

$$S = \{1, 2, 3, 4, 5, 6\}$$

A coin is tossed :

$$S = \{H, T\}$$

Combined samples

$$\{(1, H)(1, T), (2, H)(2, T), (3, H)(3, T), (4, H)(4, T), (5, H)(5, T), (6, H)(6, T)\}$$

Q7

Variable	f	cf
0-5	2	2
5-10	5	7
10-15	7	14
15-20	13	27
20-25	21	48
25-30	16	64

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) h$$

$$N = 64$$

$$N/2 = 32$$

$$= 20 + \left[\frac{32 - 27}{21} \right] \times 5$$

$$= 21.19$$

Mode

$$M = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

$$= 20 + \left(\frac{21 - 13}{42 - 13 - 16} \right) \times 5$$

$$= 23.08$$

Q8

X	-2	-1	0	1	2	3
P(X)	0.1	k	0.2	2k	0.3	k

a) Find value of k

$$\sum P(X=x) = 1$$

$$\Rightarrow 0.1 + k + 0.2 + 2k + 0.3 + k = 1$$

$$\Rightarrow 4k + 0.6 = 1$$

$$k = \frac{0.4}{4} = 0.1$$

$$\boxed{k = 0.1}$$

b)

Q9

x	0	1	2	3	4
$P(X=x)$	0.35	0.15	?	0.10	0.20

(a) $\Rightarrow \sum P(X=x) = 1$

$$= 0.35 + 0.15 + x + 0.10 + 0.20 = 1$$

$$x = 1 - 0.8$$

$$x = 0.2$$

(b) As $F(x) = P(X \leq x)$

~~for~~ $F(x) =$

$$F(x) = P(x \leq 0) = 0$$

$$F(x) = P(x \leq 0) = P(X=0) = 0.35$$

$$F(x) = P(x \leq 1) = P(X=0) + P(X=1) = 0.50$$

$$F(x) = P(x \leq 2) = P(X=0) + P(X=1) + P(X=2) = 0.70$$

$$F(x) = P(x \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3) = 0.80$$

$$F(x) = P(x \leq 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) = 1.00$$

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.35 & 0 \leq x < 1 \\ 0.50 & 1 \leq x < 2 \\ 0.70 & 2 \leq x < 3 \\ 0.80 & 3 \leq x < 4 \\ 1.00 & x \geq 4 \end{cases}$$

Q10

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

Expectation probability b/w 1 & 3 (inclusive)

$$\begin{aligned} E(X) &= 1(0.3) + 2(0.4) + 3(0.2) \\ &= 0.3 + 0.8 + 0.6 \\ &= 1.7 \end{aligned}$$

Q11

$$f(x) = \frac{1}{20}, 0 < x < 20$$

Q12

(a)

$$f(x) = kx^2 \quad 0 < x < 3$$

$$= \int_{-\infty}^{\infty} f(x) dx = 1$$

$$= \int k(x)^2 dx$$

$$= k \left[\frac{x^3}{3} \right]_0^3 = 1$$

$$= 9k = 1$$

$$\Rightarrow \boxed{k = 1/9}$$

(c) $P(1 < x < 2)$

$$= F(2) - F(1)$$

$$= \frac{1}{9} x^2$$

8/14

x	0	1	2	3	4
$P(X=x)$	0.10	0.25	0.30	0.20	0.15

(i) $E(x) = 0 \times 0.10 + 1 \times 0.25 + 2 \times 0.3 + 3 \times 0.2 + 4 \times 0.15$
 $= 0 + 0.25 + 0.60 + 0.60 + 0.60$
 $= 2.05$

(ii) $E(x)^2 = 0 \times 0.10 + 1 \times 0.25 + 0.3 \times 4 + 9 \times 0.2 + 16 \times 0.15$
 $= 0 + 0.25 + 1.2 + 1.8 + 2.4$
 $= 5.65$

(iii) $Var(x) = E(x^2) - [E(x)]^2$
 $= 5.65 - 4.20$
 $= 1.45$

Q15

$$f(x) = \begin{cases} x/8, & 0 < x < 4 \\ 0, & \text{otherwise} \end{cases}$$

(a) $\int_0^4 x/8 dx = \left[\frac{x^2}{8 \times 2} \right]_0^4 = \frac{16}{16} = 1$

$\therefore$ It is a pdf
as $\int_0^4 f(x) dx = 1$

(b) $E(x) = ?$